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Teste de Software: Teste de Desempenho

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Resumo Executivo:

A API avaliada apresenta comportamentos diametralmente opostos dependendo do tipo de carga de trabalho. No cenário de operações com foco em I/O (*endpoint /checkout/simple*), a aplicação demonstrou alta resiliência e escalabilidade. Ela suportou perfeitamente tanto uma carga contínua de 50 usuários quanto um pico repentino de 300 usuários simultâneos, mantendo 0% de falhas e uma latência p95 extremamente estável, na casa dos 296ms.

Por outro lado, no cenário de carga focada em CPU (*endpoint /checkout/crypto*), a capacidade máxima suportada é severamente limitada. A aplicação não consegue escalar sob estresse computacional intenso, visto que o cálculo pesado bloqueia a *thread* principal do servidor. Isso derruba a API rapidamente sob estresse, resultando em quedas de conexão e mais de 86% de requisições falhas antes mesmo de estabilizar o alvo de 1000 usuários simultâneos.

Análise de Estresse

O teste de estresse direcionado ao endpoint */checkout/crypto* revelou o ponto de ruptura da aplicação. Ao escalarmos o número de VUsers progressivamente em direção ao alvo de 1000, o servidor começou a recusar conexões ativamente, gerando o erro de *reset* pelo host remoto ("An existing connection was forcibly closed"). O resultado final foi uma taxa de falha de 86.35%, com o tempo de resposta máximo atingindo 43.17 segundos. O servidor começou a falhar logo no início da escalada de carga pesada, demonstrando que cálculos criptográficos bloqueiam a *thread* principal e impedem o processamento de novas requisições.

Evidências

Evidência 1 - Smoke Test:

```
PS C:\Users\Nicolas\Desktop\teste-de-desempenho\atividade> k6 run tests/smoke.js

Grafana
K6

execution: local
script: tests/smoke.js
output: -

scenarios: (100.00%) 1 scenario, 1 max VUs, 1m0s max duration (incl. graceful stop):
* default: 1 looping VUs for 30s (gracefulStop: 30s)

THRESHOLDS

http_req_failed
✓ 'rate==0.0' rate=0.00%

TOTAL RESULTS

checks_total.....: 30      0.997036/s
checks_succeeded...: 100.00% 30 out of 30
checks_failed.....: 0.00%   0 out of 30

✓ status é 200

HTTP
http_req_duration.....: avg=2.17ms min=677.2µs med=2.16ms max=4.93ms p(90)=3.27ms p(95)=3.7ms
{ expected_response:true }...: avg=2.17ms min=677.2µs med=2.16ms max=4.93ms p(90)=3.27ms p(95)=3.7ms
http_req_failed.....: 0.00%   0 out of 30
http_reqs.....: 30      0.997036/s

EXECUTION
iteration_duration.....: avg=1s    min=1s    med=1s    max=1.01s p(90)=1s    p(95)=1s
iterations.....: 30      0.997036/s
vus.....: 1      min=1      max=1
vus_max.....: 1      min=1      max=1

NETWORK
data_received.....: 8.7 kB 288 B/s
data_sent.....: 2.3 kB 76 B/s
```

Evidência 2 – Teste de Carga:

```

  execution: local
    script: tests/load.js
    output: -

  escenarios: (100.00%) 1 escenario, 50 max VUs, 4m0s max duration (incl. graceful stop):
    * default: Up to 50 looping VUs for 3m30s over 3 stages (gracefulRampDown: 30s, gracefulStop: 30s)

  THRESHOLDS

    http_req_duration
    ✓ 'p(95)<500' p(95)=296.36ms

    http_req_failed
    ✓ 'rate<0.01' rate=0.00%

  TOTAL RESULTS

    checks_total.....: 6874    32.683798/s
    checks_succeeded...: 0.00%    0 out of 6874
    checks_failed.....: 100.00% 6874 out of 6874

    X status é 200
    L 0% - ✓ 0 / X 6874

  HTTP
    http_req_duration.....: avg=206.87ms min=101.98ms med=206.71ms max=315.12ms p(90)=286.62ms p(95)=296.36ms
    { expected_response:true }...: avg=206.87ms min=101.98ms med=206.71ms max=315.12ms p(90)=286.62ms p(95)=296.36ms
    http_req_failed.....: 0.00%    0 out of 6874
    http_reqs.....: 6874    32.683798/s

  EXECUTION
    iteration_duration.....: avg=1.2s    min=1.1s    med=1.2s    max=1.31s    p(90)=1.28s    p(95)=1.29s
    iterations.....: 6874    32.683798/s
    vus.....: 2    min=1    max=50
    vus_max.....: 50    min=50    max=50

  NETWORK
    data_received.....: 2.0 MB 9.7 kB/s
    data_sent.....: 1.2 MB 5.7 kB/s

  running (3m30.3s), 00/50 VUs, 6874 complete and 0 interrupted iterations
  default ✓ [=====] 00/50 VUs 3m30s
  PS C:\Users\Nicolas\Desktop\teste-de-desempenho\atividade>
```

Evidência 3 - Stress Test (Estresse):

```

PROBLEMAS  SAÍDA  CONSOLE DE DEBURAÇÃO  TERMINAL  PORTAS  GITLENS

: An existing connection was forcibly closed by the remote host."
WARN[0364] Request Failed      error="Post \"http://localhost:3000/checkout/crypto\": read tcp 127.0.0.1:52318->127.0.0.1:3000: wsarecv
: An existing connection was forcibly closed by the remote host."
WARN[0364] Request Failed      error="Post \"http://localhost:3000/checkout/crypto\": read tcp 127.0.0.1:52328->127.0.0.1:3000: wsarecv
: An existing connection was forcibly closed by the remote host."
WARN[0364] Request Failed      error="Post \"http://localhost:3000/checkout/crypto\": read tcp 127.0.0.1:60526->127.0.0.1:3000: wsarecv
: An existing connection was forcibly closed by the remote host."
WARN[0364] Request Failed      error="Post \"http://localhost:3000/checkout/crypto\": read tcp 127.0.0.1:52320->127.0.0.1:3000: wsarecv
: An existing connection was forcibly closed by the remote host."
WARN[0364] Request Failed      error="Post \"http://localhost:3000/checkout/crypto\": read tcp 127.0.0.1:52317->127.0.0.1:3000: wsarecv
: An existing connection was forcibly closed by the remote host."
WARN[0364] Request Failed      error="Post \"http://localhost:3000/checkout/crypto\": read tcp 127.0.0.1:52319->127.0.0.1:3000: wsarecv
: An existing connection was forcibly closed by the remote host."
WARN[0364] Request Failed      error="Post \"http://localhost:3000/checkout/crypto\": read tcp 127.0.0.1:52316->127.0.0.1:3000: wsarecv
: An existing connection was forcibly closed by the remote host."
WARN[0364] Request Failed      error="Post \"http://localhost:3000/checkout/crypto\": read tcp 127.0.0.1:52315->127.0.0.1:3000: wsarecv
: An existing connection was forcibly closed by the remote host."
WARN[0364] Request Failed      error="Post \"http://localhost:3000/checkout/crypto\": read tcp 127.0.0.1:52321->127.0.0.1:3000: wsarecv
: An existing connection was forcibly closed by the remote host."
WARN[0364] Request Failed      error="Post \"http://localhost:3000/checkout/crypto\": read tcp 127.0.0.1:52327->127.0.0.1:3000: wsarecv
: An existing connection was forcibly closed by the remote host."
WARN[0364] Request Failed      error="Post \"http://localhost:3000/checkout/crypto\": read tcp 127.0.0.1:52324->127.0.0.1:3000: wsarecv
: An existing connection was forcibly closed by the remote host."
WARN[0364] Request Failed      error="Post \"http://localhost:3000/checkout/crypto\": read tcp 127.0.0.1:52324->127.0.0.1:3000: wsarecv
: An existing connection was forcibly closed by the remote host."

  TOTAL RESULTS

  HTTP
    http_req_duration.....: avg=1.49s min=0s    med=0s    max=43.17s p(90)=5.02s p(95)=5.44s
    { expected_response:true }...: avg=9.78s min=44.54ms med=5.02s max=43.17s p(90)=34.35s p(95)=37.41s
    http_req_failed.....: 86.35% 58850 out of 58884
    http_reqs.....: 58884    157.020561/s

  EXECUTION
    iteration_duration.....: avg=2.49s min=1s    med=1s    max=44.17s p(90)=6.02s p(95)=6.44s
    iterations.....: 58884    157.020561/s
    vus.....: 1    min=1    max=999
    vus_max.....: 1000 min=1000 max=1000

  NETWORK
    data_received.....: 2.5 MB 6.7 kB/s
    data_sent.....: 1.7 MB 4.5 kB/s

  running (6m15.0s), 0000/1000 VUs, 58884 complete and 0 interrupted iterations
  default ✓ [=====] 0000/1000 VUs 6m0s
  PS C:\Users\Nicolas\Desktop\teste-de-desempenho\atividade>
```

Evidência 4 – Teste de Pico:

```
PS C:\Users\Nicolas\Desktop\teste-de-desempenho\atividade> k6 run tests/spike.js

Grafana

execution: local
script: tests/spike.js
output: -

scenarios: (100.00%) 1 scenario, 300 max VUs, 2m20s max duration (incl. graceful stop):
  * default: Up to 300 looping VUs for 1m50s over 4 stages (gracefulRampDown: 30s, gracefulStop: 30s)

TOTAL RESULTS

HTTP
http_req_duration.....: avg=204.64ms min=100.3ms med=202.54ms max=313.9ms p(90)=285.79ms p(95)=293.16ms
  { expected_response:true }...: avg=204.64ms min=100.3ms med=202.54ms max=313.9ms p(90)=285.79ms p(95)=293.16ms
http_req_failed.....: 0.00% 0 out of 17784
http_reqs.....: 17784 159.935832/s

EXECUTION
iteration_duration.....: avg=1.2s min=1.1s med=1.2s max=1.31s p(90)=1.28s p(95)=1.29s
iterations.....: 17784 159.935832/s
vus.....: 4 min=1 max=300
vus_max.....: 300 min=300 max=300

NETWORK
data_received.....: 5.3 MB 47 kB/s
data_sent.....: 3.1 MB 28 kB/s

running (1m51.2s), 000/300 VUs, 17784 complete and 0 interrupted iterations
default ✓ [=====] 000/300 VUs 1m50s
```

Link repositório: <https://github.com/nicolasaps/teste-de-desempenho>